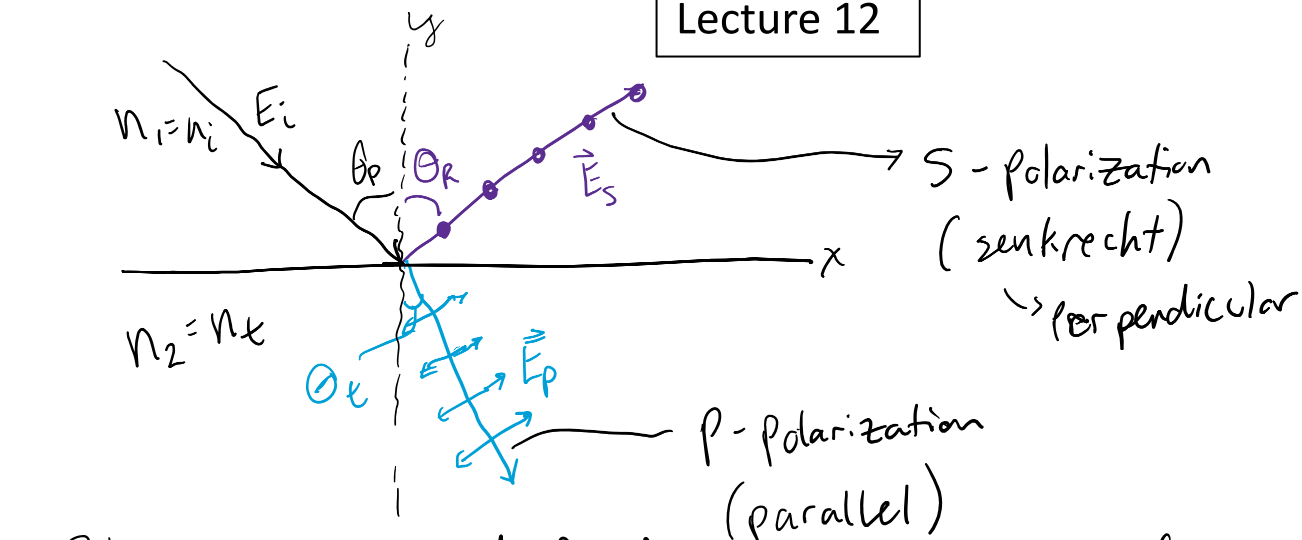
Lecture 12

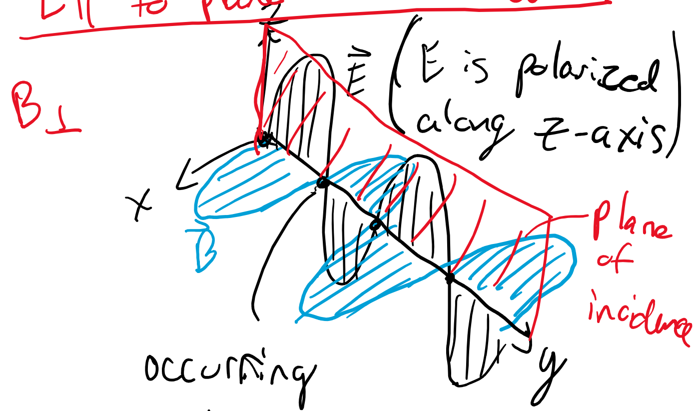


* Polarization is defined Based on E-field ↳ to plane of incidence

$E_{\parallel}$ have $B_{\perp}$; where $B_T = \left(\frac{n}{c}\right)(E_{\parallel})$; $\vec{E}_{\parallel} = \vec{E}_p$

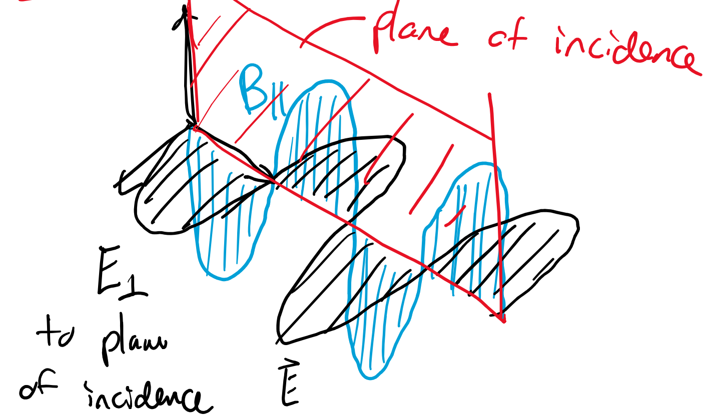
$E_{\perp} \rightarrow E_s$ has a $B_{\parallel}$; where $B_{\parallel} = \left(\frac{n}{c}\right)(E_{\perp})$

$E_{\parallel}$ to plane of incidence



occurring in phase → active power flow

$E_{\perp}$ to plane of incidence ($B_{\parallel}$)



Far-field = active power flow
"radiation"

At the surface (Boundary)

1.) $E_{tangent}$ (1) = $E_{tangent}$ (2)
2.) $B_{tangent}$ (1) = $B_{tangent}$ (2)

} EM notes

ionizing radiation → radioactive
"Not necessarily the same thing"

Using Snell's Law

$$n_i \sin\theta_i = n_t \sin\theta_t$$

$$\sin\theta_i = \frac{n_t}{n_i}\sin\theta_t$$

$$\boxed{\frac{n_t}{n_i} = (\tilde{n})} \longrightarrow \boxed{\tan(\theta_p) = \tilde{n}}$$

Fresnel's Reflection Coefficients (generalized)

$$r_\perp = \frac{E_{r\perp}}{E_{i\perp}} = \frac{\cos\theta_i - [(\tilde{n})^2 - \sin^2\theta_i]^{1/2}}{\cos\theta_i + [(\tilde{n})^2 - \sin^2\theta_i]^{1/2}}$$

$$r_\parallel = \frac{E_{r\parallel}}{E_{i\parallel}} = \frac{[(\tilde{n})^2 - \sin^2\theta_i]^{1/2} - (\tilde{n})^2\cos\theta_i}{[(\tilde{n})^2 - \sin^2\theta_i]^{1/2} + (\tilde{n})^2\cos\theta_i}$$

Fresnel's Transmissions Coefficients

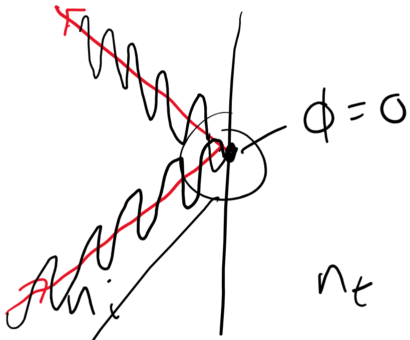
$$t_\perp = \frac{E_{t\perp}}{E_{i\perp}} = \frac{2\cos\theta_i}{\cos\theta_i + [(\tilde{n})^2 - \sin^2\theta_i]^{1/2}}$$

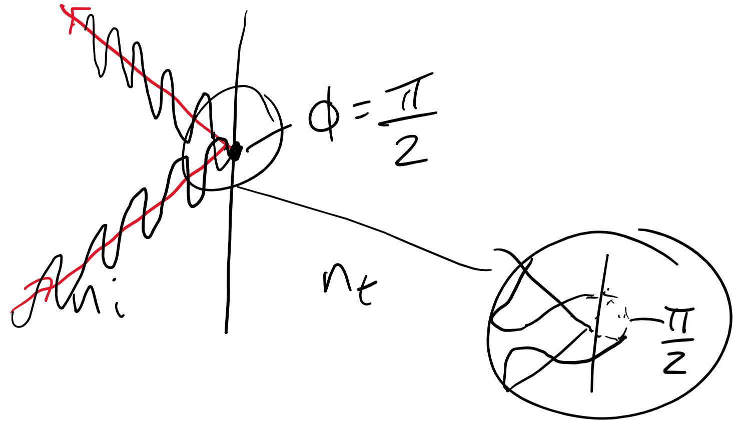
$$t_\parallel = \frac{E_{t\parallel}}{E_{i\parallel}} = \frac{2\tilde{n}\cos\theta_i}{(\tilde{n})^2\cos\theta_i + [(\tilde{n})^2 - \sin^2\theta_i]^{1/2}}$$

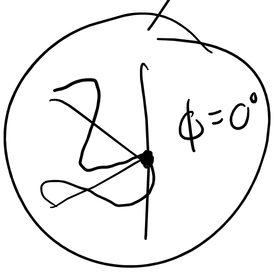
$\boxed{r_\parallel + (\tilde{n})(t_\parallel) = 1}$ and $\boxed{r_\perp + 1 = t_\perp}$

In general

$$r_\perp = |r_\perp| e^{-j\phi_\perp}$$ ← spatial ocsillation angle







* transmission and reflection coefficents can be used to see what the phase is at the boundary between n_i and n_t